

RAJESH NAUTIYAL

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PROFESSIONAL SUMMARY

Automation and Regression Lead with 14+ years of experience in network automation testing, QA automation, regression validation, and infrastructure management for Cisco enterprise networking platforms. Experienced in Cisco NXOS, IOS-XE, Python, PyATS, TCL, Jenkins job management, Linux administration, Cisco UCS, VMware ESXi, AWS, and automation lab ownership. Led regression ecosystems involving 150+ devices, 500+ automation scripts, and multiple parallel release validation cycles. Built an AI-assisted triage solution that reduced failure triage time from 1-2 hours to approximately 15 minutes by analyzing logs, Jenkins failures, device outputs, test results, and historical defect data.

TECHNICAL SKILLS

Network Platforms: Cisco NXOS, IOS-XE, N7K, N9K, N9Kv, ASR1K, CAT9K, CSR1000v

Automation and Scripting: Python, PyATS, TCL, Bash, Shell Scripting, YAML, JSON

Python Libraries and APIs: requests, pandas, paramiko, netmiko, pytest, REST APIs, Cisco internal automation libraries

Network Programmability Testing: YANG Models, REST APIs, Model-Driven Programmability Validation, Device Configuration Validation, Telemetry Validation, API Response Validation

CI/CD and Tools: Jenkins job execution and management, Git, CVS, Visual Studio Code, GitHub Copilot, Codex

Infrastructure and Cloud: Linux RHEL7/RHEL8, Cisco UCS, VMware ESXi, AWS

Testing and QA: Regression Testing, Release Validation, Automation Validation, L2/L3 Protocol Script Testing, Failure Analysis, Log Analysis, Root Cause Support, Defect Tracking

Protocol Script Test Coverage: VXLAN, BGP, OSPF, EIGRP, HSRP, VRRP, VLAN, STP, EVPN, MPLS

PROFESSIONAL EXPERIENCE

CAPGEMINI | Dec 2022 - Present

Role: Senior Professional II | Regression and Automation Lead | Project: Cisco

- Lead regression and automation activities for Cisco NXOS and IOS-XE platforms while working directly with Cisco engineering teams.
- Lead a team of 5 engineers responsible for regression execution, automation maintenance, failure triage, lab readiness, and release validation, regression analysis, defect follow-up, and quality sign-off for enterprise networking releases.
- Manage automation infrastructure involving 150+ physical and virtual devices across Cisco enterprise networking platforms.
- Own automation lab readiness, topology creation, hardware allocation, Linux server administration, and Python environment management.
- Manage and maintain 500+ regression automation scripts for continuous validation across multiple parallel releases.
- Run and manage Jenkins-based regression jobs for scheduled and on-demand automation execution.
- Perform release validation, - Developed an AI-assisted triage bot to analyze Jenkins failures, traceback errors, device logs, test result files, and historical defect data.
- Designed bot instructions to identify probable root causes, suggest next debugging steps, and match current failures with past known issues.
- Reduced regression failure triage time from 1-2 hours to approximately 15 minutes through AI-assisted analysis.
- Trained 10+ trainees and junior engineers on automation workflows, regression execution, and Cisco platform debugging.

TATA CONSULTANCY SERVICES (TCS) | Mar 2012 - Dec 2022

Role: Associate Consultant | Technical Lead | Mar 2018 - Dec 2022 | Project: Cisco

- Led automation and regression testing initiatives for Cisco NXOS and IOS-XE platforms.
- Tested, validated, and supported Python-based automation scripts using PyATS, YANG models, REST APIs, JSON/YAML parsing, and Cisco internal automation libraries.
- Set up environments and automation flows for model-driven programmability validation, including device configuration validation, telemetry validation, and API response validation.
- Managed automation labs, Linux servers, regression infrastructure, and Cisco UCS-based test environments.
- Supported regression testing and automation validation for protocol scripts covering VXLAN, BGP, OSPF, EIGRP, HSRP, VRRP, VLAN, STP, EVPN, and MPLS.
- Performed failure analysis, root cause support, defect tracking, and quality validation for Cisco platform releases.
- Coordinated automation integrations and release validation activities across multiple Cisco platform teams.

Role: Automation Testing Engineer | May 2012 - Feb 2018 | Project: Cisco

- Developed TCL automation scripts for router and switch command validation.
- Performed regression testing and debugging support for Cisco NXOS and IOS-XE platforms.
- Worked on L2/L3 protocol script regression testing and automation validation.
- Analyzed device logs, test failures, automation errors, and regression results to identify issues.
- Supported defect reporting and issue tracking using Cisco internal defect management tools.

SELECTED PROJECTS

AI-Assisted Regression Failure Triage Bot: Developed an AI-assisted triage bot to analyze Jenkins failures, traceback errors, device logs, regression result files, and historical defect data. The bot identifies probable root causes, suggests next debugging steps, and matches current failures with past known issues.

Cisco Regression Automation Framework: Managed a regression automation ecosystem supporting 150+ devices and 500+ automation scripts across Cisco NXOS and IOS-XE platforms, including lab readiness, Jenkins job execution, Python environments, and continuous regression workflows.

YANG/REST API-Based Automation Testing: Tested and validated Python automation scripts for Cisco network programmability use cases using YANG models, REST APIs, JSON/YAML parsing, and Cisco internal libraries. Set up test environments and automation flows for device configuration, telemetry, and API response validation.

KEY ACHIEVEMENTS

- Managed automation infrastructure with 150+ devices and 500+ regression automation scripts.
- Completed 300+ automation integration activities.
- Reduced regression failure triage time from 1-2 hours to approximately 15 minutes using AI-assisted analysis.
- Trained 10+ engineers on automation and regression workflows.
- Certificate of Excellence - TCL to PyATS Migration; 4x Special Initiative Award Recipient; Star of the Month Award.

EDUCATION AND CERTIFICATION

Education: Bachelor of Technology (B.Tech), Information Technology - GB Pant University of Agriculture and Technology, 2011

Certification: AWS Certified Developer Associate, 2022